Smart IOT System for Monitoring and Controlling Livestock Parameters

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Abstract: In this paper we consider the products like quality and welfare of animals which are closely related to the environment parameters in livestock building. A monitoring and control method of environmental parameters in livestock buildings based on wireless sensor network is proposed in this system. The parameters like temperature, humidity, light intensity, carbon dioxide concentration, ammonia concentration, and hydrogen sulfide concentration can be monitored and controlled in real time by this method. These parameters can be adjusted and controlled through WLS algorithm and also can minimize the deviation criteria. Along with this, the system can also track the location of the animals in the livestock building using several tracking nodes placed around the neck of the animal. In addition, the system proposes the ALEXA based voice assistant which is a user friendly system that can assist us to control and monitor the livestock parameters. Compared with the traditional method, the proposed method reduces the high labour cost and energy consumption will be decreased.

Keywords: Wireless sensor Network, Tracking, ALEXA, GPS, GSM, Zigbee, MYSQL.

1. Introduction

From [1] Livestock husbandry is also an important sector which place a key role in economy employment. Recently most of the livestock forms are get effected by the environmental parameters such as temperature, humidity, light intensity, Carbon dioxide(CO_2), Ammonia(NH_3), Hydrogen sulfide(H_2S). The environmental condition of the poultry house and livestock stall has become an important factor in the development of poultry house and livestock production. In the process of poultry and livestock breeding, controlling the poultry environment has an important significance to the animal health growth to maximize the use of its growth performance. The proposed system of livestock and poultry production have shown that the growth, health, reproduction status, and forage utilization of livestock and poultry are restricted by the environment its house and stall.

According to The Latest Agriculture Industry Standard of India, it is the main environment parameters in livestock building such as temperature, humidity, light, carbon dioxide concentration, ammonia concentration, and hydrogen sulfide concentration. In [2], Most of the research results, at present are only in the theoretical stage and the experimental stage, few of them have been tested for a long time, and the accuracy and stability of data acquisition and system still cannot be verified before having mass field testing in livestock building.

From [3] for improving the management of livestock husbandry, there is a need for tracking the location of animals in livestock building using an automated system. Since the animals are in the closed environment but still there may be a chance of missing of animals or may be a chance of theft of animals by some other persons. Therefore livestock farm may need this tracking system. The Range will be created according to the physical area of the livestock building.

For user friendly, AMAZON ALEXA was introduced in this system. According to the commands given by the user, it will control and monitor the livestock parameter in the real time.

The rest of the paper deals with the following concepts

Related work--gives a comparison between proposed systems with other References.

Measuring Parameters--deals with the parameters that are going to measure and also about the nodes that are being placed in the livestock Farm.

Proposed System--deals with the Pictorial Representation of the system with proper description.

Algorithm and Flowchart—deals with the WLS algorithm and it's Flowchart with proper required equations.

Conclusion—it represents what the proposed system finally achieved compare with Past systems.

If any animal crosses that range, alertness will be obtained to the user in the form of alarm and Message to mobile with the help of GSM and controller.

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2. Related Work

Let us consider [1] i,e. Yu Zhang's Environmental Parameters control Based on Wireless Sensor Network in Livestock Buildings, Monitoring and controlling is done automatically using server. But, estimation of environmental Parameters are not done. Similarly in Addition to this we Proposes the Tracking of cattle in Livestock building. Along with this we Includes the Amazon's Alexa Voice Assistant system for better Accessing and Controlling of Parameters by the User.

Similarly consider the [2] i,e. WenJie Tian's Environmental Monitoring system of Household Security Robot Based on Wireless Mesh Network, Monitoring is only achieved using Wireless mesh Network. From [3] i,e Sung-Gi Choi's Design and Implimentation of a GPS-enables Mobile Wireless sensor Network for Livestock Herd Tracking in Mangolian Nomadic Herding, Tracking of animal location is achieved.

Table 1: Comaprison between proposed system and others

PARAMETERS	REFERENCE1	REFERENCE2	REFERENCE3	PROPOSED SYSTEM
Sensors	Temperature Humidity,light Illumination, Gas concentration Sensor.	Temperature, Distance sensor,video camera.	_____	Temperature, Humidity,ammonia, CO_2, H_2S,light Illumination sensor.
PROCESSING UNIT	_____	ARM920T	Normal MCU unit	ARM cortex-A53
COMMUNICATION MODULE	Zigbee	Zigbee	Zigbee	Zigbee
PROTOCOLS	http protocol	HWMP, http Protocol.	TCP/IP	TCP /IP, http protocol
PLATFORM	_____	Embedded C	PYTHON	PYTHON V3.7,Embedded C
SERVER	SQL server	Web Server	Web Server(Apache)	MYSQL server
DATABASE	SQL data base	____	MYSQL database	MYSQL data base

3. Measuring Parameters

3.1 Air temperature

From [1], the sunlight varies with the seasons and time, so temperature changes, in the livestock building is not set in stone. When temperature changes, livestock will react accordingly. Take the temperature dropping, for example; livestock will make spontaneous activity weakened decrease sweat glands respiratory

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activity, in order to reduce the heat loss. At the same time, under the cold condition, livestock and also use the oxidation of nutrients in the body to increase the body's heat peroxide and maintain body temperature regulation.

Temperature sensor makes use of various physical properties with temperature change law of the category of sensor temperature conversion into electricity. According to the principle of work, it can be divided into pyroelectric temperature sensor, thermal resistance temperature sensor, and PN junction temperature sensor. Surface temperature is the important basis of animal epidemic prevention. As per the Indian standards 34.5 to 41C is the standard temperature range.

3.2 Relative Humidity

In [1], it is the most direct impact of air humidity on livestock, which affect livestock thermal regulation. Under the condition of high temperature, livestock regulate their own body temperature by evaporative heat, but high humidity can reduce the temperature difference between skin and air difference to reduce the heat. Humidity is an environmental factor and promotes fattening and growth of livestock with other environmental factors together. The rise of humidity will have a harmful effect on the composition of the breast milk.

Common method to measure humidity is absolute humidity and relative humidity. According to the different devices, the humidity sensor can be divided into two major categories of the resistive one and the capacitive one. It is an important environmental parameters index, and humidity sensors are widely used in green houses and livestock breeding places to monitor the humidity. The standard Relative humidity in India for livestock Farms are ranges from 30% to 60%.

3.3 Ammonia

Ammonia is absorbed by livestock, when the quantity is small, it can be directly transformed into urea and be eliminated from the body, but it also affect the disease resistance of livestock. From [1], if the quantity is large, blood can bring it through the lungs and generate the low iron into the alkaline high iron red element, which influences blood oxygen capacity of livestock, causing symptoms such as anemia and lack of oxygen.

3.4 Hydrogen sulfide

Hydrogen sulfide enters into the livestock by cattle respiratory system. Combined with sodium ions, it can produce sodium sulfide, in respiratory mucosa. Which stimulates mucous membrane, causing inflammation, and even tissue edema.

In [1], Hydrogen sulfide is harmful for the circulation of the blood system. It enters into livestock by alveolar. Finally it affects cell oxidation ability, causing systemic poisoning symptoms of animals.

3.5 Carbon Dioxide

Most of the carbon dioxide comes from animals breathing activity. It is nontoxic substances if carbon dioxide is too much, the concentration of oxygen in the air will drop greatly. Thus inhibit normal breathing and physiological metabolism of livestock.

3.6 Light intensity

Light intensity is the luminous flux in the form of object area and surface, it is indispensable to livestock environmental factor, which plays a key role in the process of external conditions, such as survival, growth, development, and breeding. On the natural conditions, natural light has a high regularity, with the seasons changing, day and night shifting. It makes a lot of functions of livestock and poultry body deal with regular sunlight life activities, such as sleep, tissue metabolism, hormone levels, and immune and nervous system.

In [1], Light intensity sensor is a part of the photoelectric sensor, which can take advantage of the component of the photoelectric effect to convert light flux into electricity. Light is closely related to the physiological cycle regulation of livestock, so the light sensor is widely used in the field of monitoring and control of agricultural production.

3.7 Harmful gas transducer

From [1], Gas sensor is a type of sensor that can convert gas concentration measurement to the amount of voltage or electric current. According to the different principle, gas sensor can be divided into semiconductor gas sensors, electro chemical gas sensors and infrared gas sensor.

Among these, semiconductor gas sensor takes advantage of high sensitivity, fast response speed, good maintainability and low cost. Harmful gases will directly affect the quality of growth and livestock products. Thus most of the harmful gas sensor are used in livestock environment intelligent control field.

Table 2: Type of parameters

Parameters	definition	type	Power supply
Temperature sensor	Temperature sensor is a device, which is used to sense the heat of temperature	digital	3.5V to 5.5 V
Humidity sensor	It is the sensor which measure the amount of water vapour in the surrounding air.	digital	3.5 V to 5 .5V
Light sensor	It is the sensor which measures the light intensity in the surrounding environment.	analog	2 V to 5 V
Ammonia sensor	It is the gas sensor which measures the amount of ammonia present in the air.	digital	3 V to 5.0 V
Co_2 sensor	It is the gas sensor which measures the amount of carbon dioxide present in the air.	digital	3 V to 5.0V
H_2S sensor	It is the gas sensor which measures the amount of hydrogen sufide in the air.	digital	2 V to 5.0

3.8 Wireless sensor network

WSN is a new type of information awareness and network system of collection. It can obtain all kinds of detailed and accurate target data information at any time, any place, and any environment too.

From [2] and [16] Several Sensor nodes are placed at different places around the animals in the Livestock building. These sensor nodes are connected to the master node through the Zigbee protocol (IEEE 802.15.4). The Master node in WSN was equipped with GPRS module, which can send the data via mobile communication network to the Controller.

3.9 WSN for Tracking

From [3] and [15] and for tracking system, the sensor node was equipped with Zigbee module (IEEE 802.15.4), Micro controller unit module and power supply. Where as the master node was equipped with an additional block device called GPS module. Zigbee is the protocol used between the sensor node and master node where as GPRS is for master node and controller.

GSM module connected to the master tracking node to give proper alertness to the user. Each animal in the livestock building was allocated with a unique ID which creates flexibility to identify the cattle in the livestock. The type of cattle is also identified using ID number. The coordinates of latitudes and longitudes at the animal location is sent to the mobile through GSM module.

3.10 Amazon Alexa

Simply known as Alexa, is a vitual assistant AI developed by Amazon. It can control several smart devices as per the requirements by the user. From [4], most devices with Alexa allow users to activate the device using a wake word. It allows a user to say a command, such as "Alexa" or "Alexa wake". Currently, interaction and communication with Alexa are available only in English, German, French, Italian, Spanish, Portuguese, Japanese, and Hindi. The code almost entirely in the cloud, using Amazon's AWS LAMBDA service.

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The format of common NLP (natural language process) contents wake work, invocation name and utterances. Wake work indicates the device coming to the wake (active) from sleep mode. Invocation name indicates skill identifier i.e. name of system. An utterance indicates the way of conversation.

4. Proposed System

The figure 4.1 represents the way of arranging the sensor nodes to different corners in livestock building. These Sensor nodes are connected to the Center node using Zigbee protocol.

Similar to this, Several Tracking nodes are connected to the Tracking sink node using same zigbee Protocol.

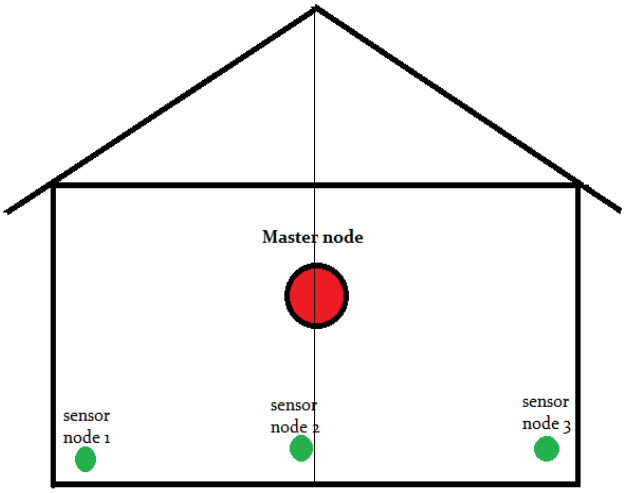

Fig. 1: Position of sensor nodes installed

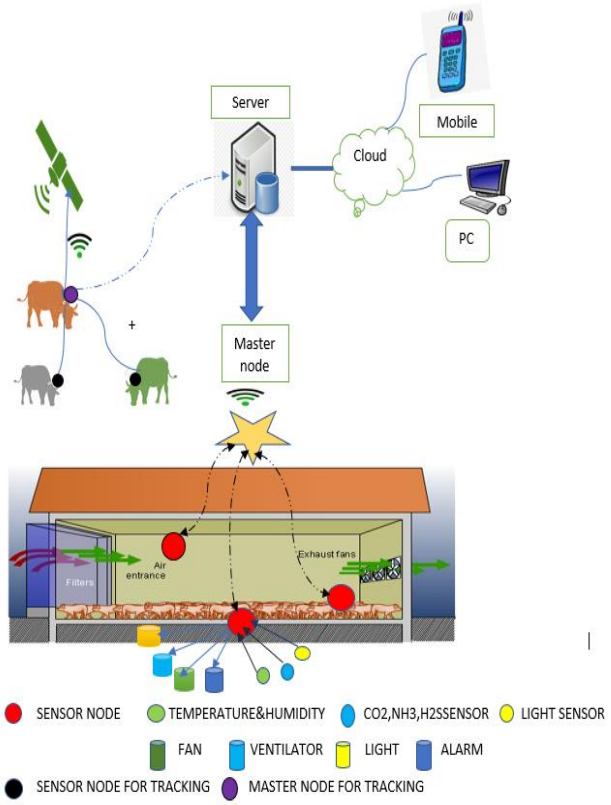

Fig. 2: Overall system Design

The proposed system consists of monitoring, controlling and tracking functions as shown in Figure 4.2. Several sensor nodes are placed at different places in the livestock building. Each sensor node contains sensors such as temperature, humidity, light illumination and gas sensor and each sensor node is connected with different actuators like fan, light, ventilator, alarm etc. Master node contains all the information which were coming from different sensor nodes through zigbee protocol (IEEE 802.15.4). From master node, the information goes to back end server.

By using WLS algorithm, the code is dumped in to the server to attain proper control over the actuators. From [12], if any parameter exceeds the standard range, then the actuators will control the deviation until it has been set.

Similarly, tracking nodes are attached to animals by means of a belt. These nodes are connected to a master node via zigbee protocol (IEEE 802.15.4) which contains multiple hardware components like GPS/Cellular module, ZigBee module, MCU module, power supply module. Here by using GPRS protocol the master node is connected with server. If any animal escapes from the given range, then proper vigilant is generated to the mobile containing the latitude and longitudinal values of that particular locations. According to the area of livestock building, the range is allocated by the user.

4.1 Monitoring and Controlling Using Alexa

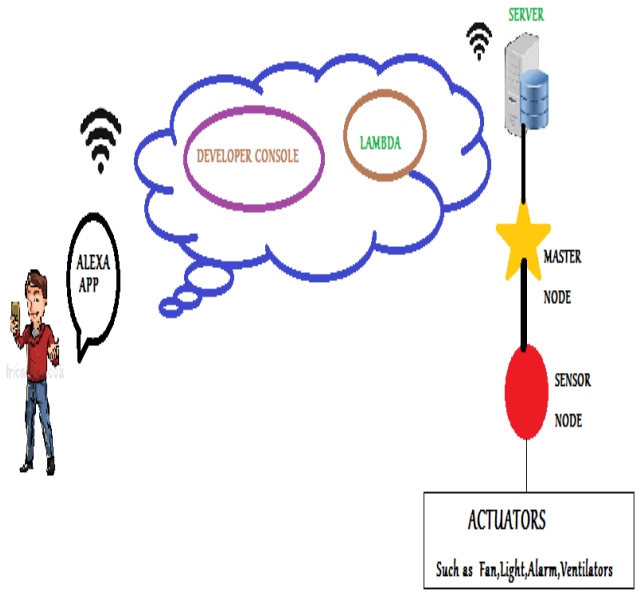

Fig. 3: Working of an Alexa system

The figure3 represents monitoring and controlling environment parameters using amazon ALEXA kit. The commands from us carried to the AWS (Amazon Web service) Cloud through the Amazon Alexa APP. Alexa Device will converts the NLP (Natural Language Process) into Machine Language. Developers Console and LAMBDA Tools are Present in the AWS cloud. Developer Console Provides a Streamlined Experience to help you create, manage, and Publish skills. The following are the Steps that leads you to become an Alexa Developer

1. Design Your Voice Experience
2. Build Your Voice USER Interface in the Alexa Developer console.
3. Build Your Backend to power Your Skill.
4. Test and optimize.
5. Submit, Get Certified, and Publish.

In AWS, Lambda application is a Combination of Lambda functions, event sources, and Other Resources that work together to Perform Tasks. All you need to do, for a custom Alexa Skill, is to upload your code into a Lambda function, which will execute in response to Alexa Voice Interactions. Currently Lambda Supports Node.js, Python, Java, and C#.S

Lambda is Connected to the Server via IOT application and then to the Sensor node. Hence controlling and monitoring is done.

5. Algorithm and Flowchart

Along with monitoring, the controlling and estimating of environmental parameters can be done by using WLS algorithm. The method of weighted least square can be used when the ordinary least square assumption of constant variance in the error is violated is called Heteroscedasticity. The model under consideration is

$$Y = X\beta + \epsilon$$

Where, Y = dependent variable

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X = independent variable
β = estimation of parameter
€ = error

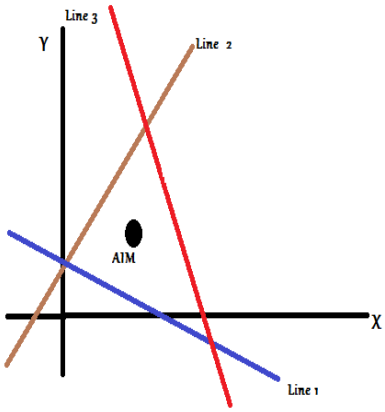

Fig. 4: Multi sensor perceptual information vector equivalent

Where now € is assumed is to be (multi vibrate) normally distributed with mean vector zero and non-constant variance-covariance matrix

$$\begin{pmatrix} \sigma_1^2 & 0 & - & - & 0 \\ 0 & \sigma_2^2 & - & - & 0 \\ - & - & - & - & 0 \\ 0 & 0 & - & - & \sigma_n^2 \end{pmatrix}$$

If we define the reciprocal of each variance σ_1^2, as the weight $w_{1=1/}$ σ_1^2, then let matrix w be a diagonal matrix containing these weights

$$w = \begin{pmatrix} w_1 & 0 & - & - & 0 \\ 0 & w_2 & - & - & 0 \\ - & - & - & - & 0 \\ 0 & 0 & - & - & w_n \end{pmatrix}$$

The weighted least square estimation is then

$$\beta_{wls=\arg\beta_{min}} \sum_{i=1}^{n} \varepsilon_i^{-x^2}$$

Or

$$(x^T\ w\ x)^{-1}\ x^T\ w\ y$$

Residual sum of square

$$R_{ss}(\beta) = \sum_{i=0}^{n} (y_i - x_i * \beta)^2$$

The minimizing deviation criteria of environment parameters control. The control of temperature and humidity it has been a significant problem. Many people attempt to find out an accurate and efficiency way to control temperature and humidity.in order to solve this problem, a useful method will give us some aspiration.

Environment parameters influence each other nonlinearly so that they cannot be adjusted separately. Set S_i and S_i' as standard values and measured values of the monitoring parameters. Define the minimizing deviation criteria of environment parameters control for

$$Q = \sqrt{\sum_{i=1}^{n} \left(\frac{S_i' - S_i}{S_i}\right)^2}$$

If Q tends to be the minimum, the control algorithm works. In practice, usually set the temperature and humidity as a group, and set the gas as the other group.

The basic steps that are involved in the controlling process are as follows

1. Monitor the actual state Parameters.
2. Calculation of Actual Measurements.
3. Compare the actual parameter values with the fixed range.
4. If the obtained difference was large, then automatically controlling action will be takes place along with proper alertness.
5. The controlling still done until the difference was too small.

Internally the WLS algorithm Estimates the future Parameters values by taking the past values. The Estimation values may not be exact. There may be a chance to get slight variation between estimated value and real value.

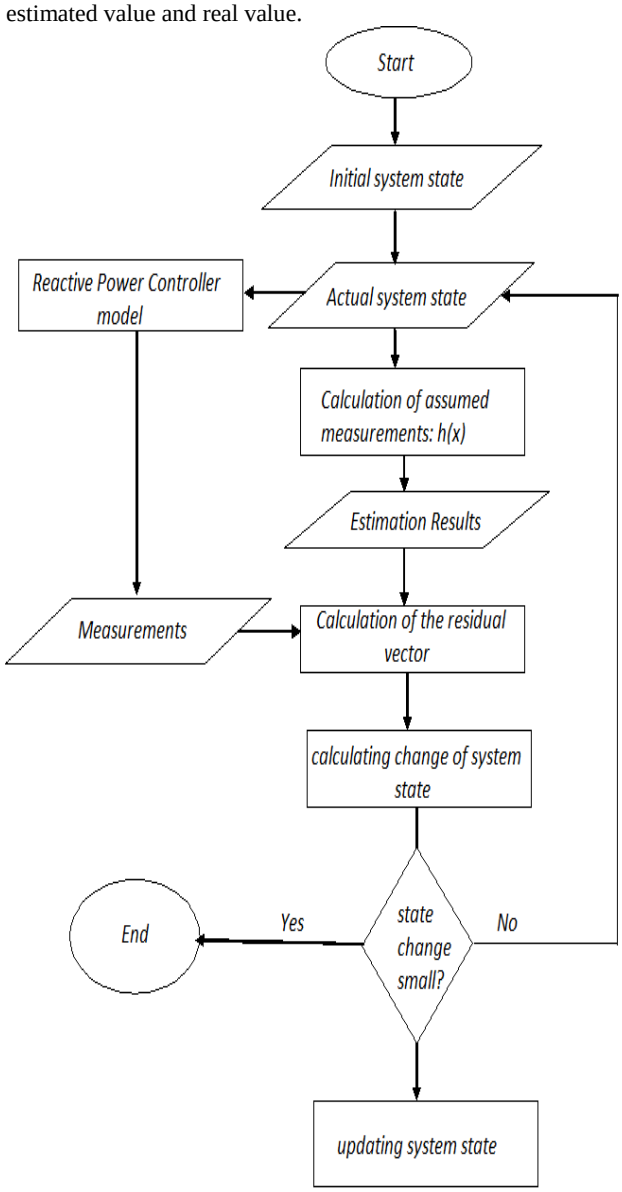

Fig. 5: Flow chart

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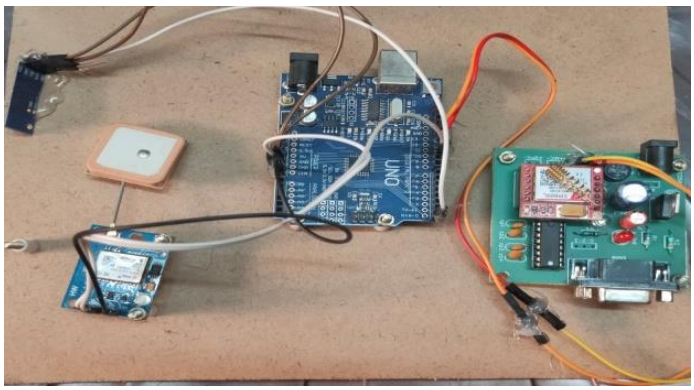

Fig. 6: Tracking Sink (master) node

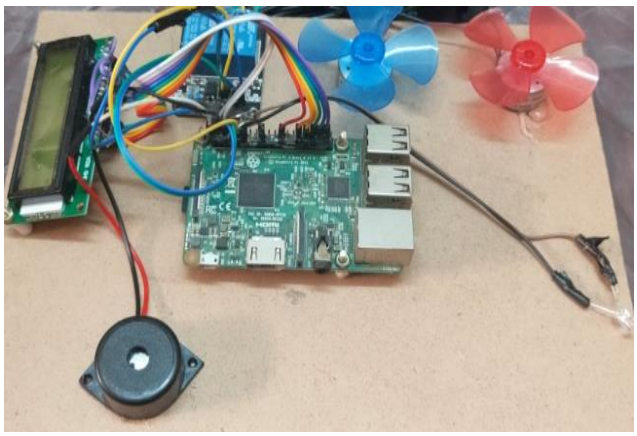

Fig.7: Controlling unit

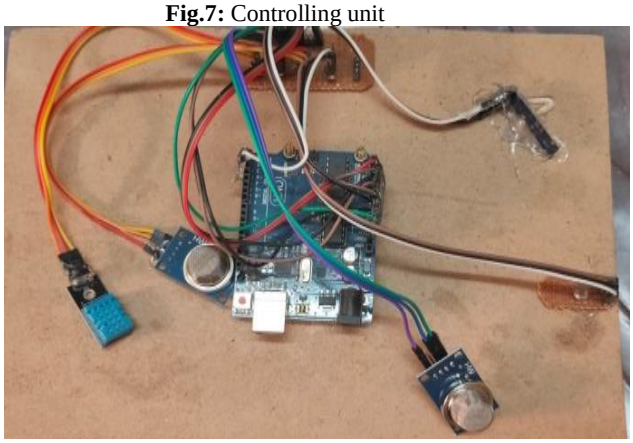

Figure 8 Sensor node

6. Conclusions

In this paper, we presented the smart IOT system for monitoring and controlling livestock parameters the entire system is designed using ARM processor embedded single board with 1GHz processing speed i.e. raspberry pi. Overall several sensor node are created to convert physical data into electrical quantity. The recent advantage in IOT has several benefits and advantages over manual wired/ wireless system. For better management of livestock building, tracking application plays a major key role. In additional, we added ALEXA voice assistant system which makes the controlling and monitoring easily by the user.

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